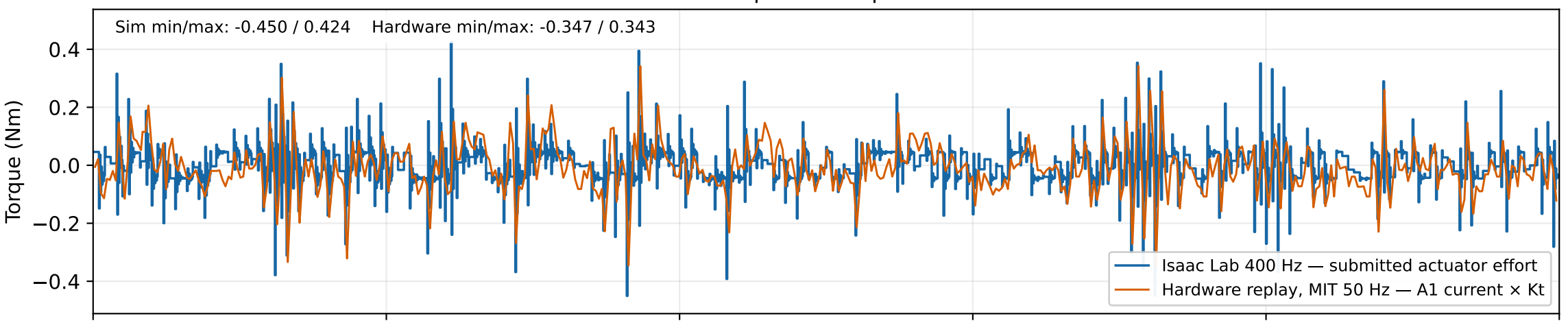
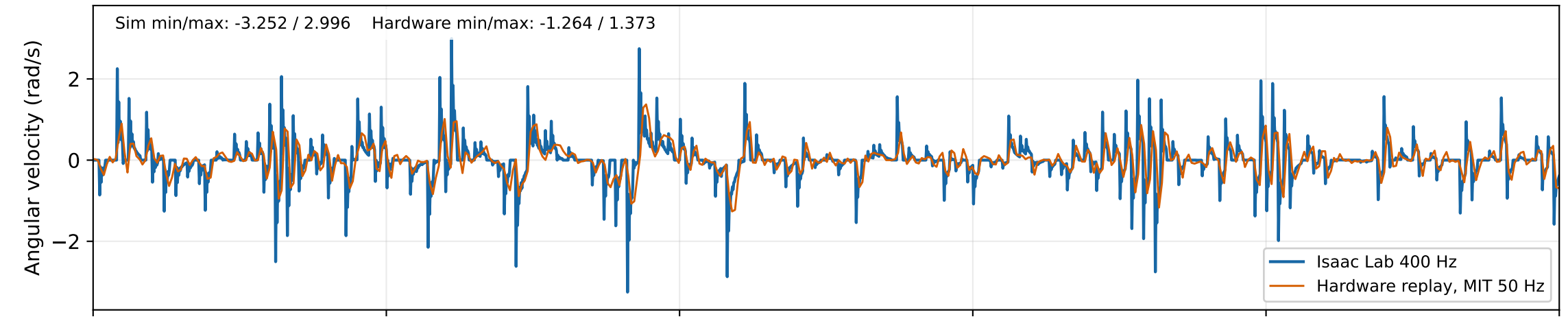
Steering motor: simulation versus 50 Hz hardware replay

Same 50 Hz-held model_2500 position targets · source episode 12.5–17.5 s
Airborne corrected-zero MIT replay versus saved nominal fixed-base simulation · no time shift

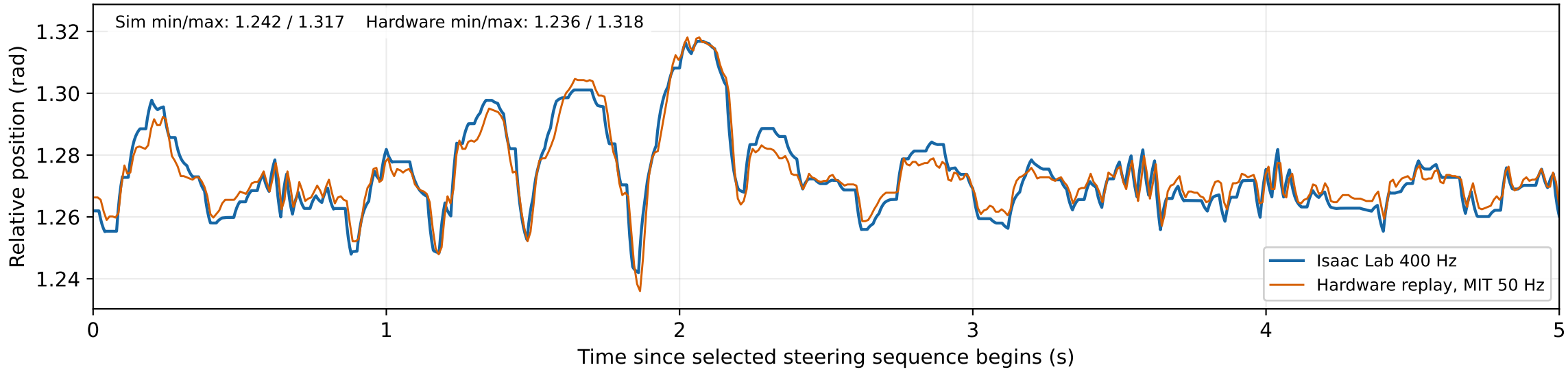
Actuator torque before passive friction



Steering angular velocity



Steering angle from the fixed straight reference



Torque is simulation effort versus current-derived hardware estimate, not net torque after losses.
Position retains each run's verified straight zero; nominal Kp/Kd = 10/0.2. Support and actuator dynamics are not fully matched.